

Karl Friston & Alexander Ororbia: "First Day ~ Symposium Roundtable Dis

Symposium Roundtable Discussion" | 57:53

is

Carl hello welcome back

everyone I'm here with Alex

oria we may have Carl join we may have

someone else join if you want to join

maybe write it in the YouTube live chat

and maybe I'll email you a link

otherwise write some

questions however to kick things off

Alex feel free to say hello and start in

and I know that there's things that we

can explore together sure you just

picked it back to me and I was saying to

you all right well I I'll just introduce

myself because I'm not entirely sure how

most of the audience if they know of who

I am so I'm Alex Ria I am a

computational neuroscientist and

cognitive scientist who has worked for

active inactive inference and

specifically predictive coding versions

of it for quite some time I'm Carl's

friend and collaborator uh some of you

might be aware of more of our recent

arguments of mortal computation which is

a Cory or a new coraly or another coraly

of the free energy principle and uh yeah

and you know I'm all about what's wrong

with deep learning in machine

intelligence and I'm usually the guy you

go to if you want to attack back

propagation of Errors large language

models Transformers kind of relates to a

lot of the discussion I was able to

catch Snippets of earlier today so if

you want me to help dismantle what's
what's the problem with deep
learning you know that that can be fun
too so that at least if nothing else
everyone now knows who I am Carl needs
no
introduction maybe start off with that
while I try to figure out why he cannot
join oh okay uh what part of that do you
want me
to I mean I can go into Manifesto but
I'm not sure that's what this panel
should be about
out maybe beginning with a five minute
Manifesto and then there will be some
questions and uh we'll see where we
get well maybe to try to make it useful
to the audience uh for the moment
although I wasn't intending the panel to
be focused about that um maybe one thing
that we can talk about is there was this
discussion during the active inference
Workshop about metabolism and sort of
instill a homeostatic metabolic drive a
survival almost orientation to machine
intelligence and so uh Carl and I have
now been arguing for probably now about
more than a year since uh our paper came
out like on yeah around November
actually start of November talking about
uh another way to redesign reformulate
to use Ma's earlier uh uh wording uh
intelligence itself uh particularly
machine intelligence
and so mortal computation without again
if for those that might not be familiar
actually my friend Jeff also like two
years ago introduced this phrase the

idea that computation or intelligent calculations and substrate uh in biological natural systems are entangled and it's very hard to divorce those two concepts uh whereas computer science and a lot of like what cat GPT is or large language models or your favorite GPT structure is Immortal which is the idea that we can copy that program including its weight tensors and it's encoded knowledge and place it on GPU server you want and it'll behave roughly the same there's nothing inherent about its physical instantiation uh in this universe uh that dictates or shapes or uh necessarily embodies it right and so there's this like move to uh embodiment by the way active in the free energy principle has all of these characteristics mortal computation which is again as the concept I explain earlier Carl and I sort of uh recast it we strongly redid it and said that um uh because while Jeff didn't want to necessarily directly explore it we put it as it's just basically thinking of the free energy principle and taking it to its natural extreme conclusion which is that machine intelligence systems uh because they are inherently bound to their substrate must focus on this survival objective right and the free energy principle and one of its ways to talk about it uh basically is saying you know we have a markup blanket or a nested system of markup blankets and our goal in this universe is to protect

those internal States uh yet interact vicariously with our external States and so that eventually gives us our ultimate goal which is to prevent ourselves from disintegration into a heat bath it's a very physics uh form of thinking about identity and identity construction um but this is kind of what's centrally missing in basically all of machine intelligence um is this the completely right path to go there's plenty of criticisms and I'm not one to say that there's only one direction we need to go if we ever want something like artificial general intelligence um but this might be a path worth exploring it also addresses the Energy Efficiency arguments I was hearing earlier discussed uh Transformers particular I in large language models are uh grossly negligent of their energy costs they often times have the carbon footprint of a large city and that's actually now out of date as you know Carl and I wrote that about a year ago um there's even discussions I find kind of humorous by individuals I'm not going to particularly name saying we should be powering now uh these Transformers and large language models with nuclear power plants and so we're kind of going down the rabbit hole of let's just burn as much of our resources as we can to just get this particular type of intelligence I argue very strongly narrow AI uh and we're cheating ourselves of looking and embracing actually Energy Efficiency uh which

would lead us towards the real grand goal of green AI uh because clal GL global climate change is clearly and empirically a thing and we should be a little worried about that so uh mortal computation sort of helps steer everything into sort of what I would argue hopefully maybe in multiple years time the emergence of a whole new field because I consider at least my personal take I'd love to have you know heard Carl wherever he might be um but I Envision actually mortal computation is not just like the framework and definitions that Carl and I talked about which kind of take free energy uh and active inference to the strongest conclusion to a naturalistic machine intelligence maybe perhaps a new domain of thought of biomedic intelligence and where everyone else can come together and play in this really interesting space and there's a lot to unpack there but I don't think we want to make this about that uh there's plenty of talks I've given including active inference Institute we've had a podcast actually I think it was right when the paper came out so for those that are interested I just recommend you to go to the YouTube channel and watch wonderful videos that Dan LED oh is there is there a new link okay sure was was I just blabbing there to no one no that was perfect we we we just hot swapped okay I was trying to buy Carl welcome Carl we talking about

you

okay thank you for the overview on
Mortal computation and and and it was a
highlight of the

year um Carl with the evening Sunset

now how um goes it

wait I don't hear you oh oh continue

sorry it's all good Carl

continue can you hear me now yep

excellent yeah I'm sorry about that I

got locked out I was

uh paying the price for enjoying myself

for the first two

hours right I'll

wait just I really enjoyed the um the

intervening sessions um and look forward

to of comenting some the cross cutting

scen yes okay um Alex

mute it's a little noisier on jitsy and

then uh corl um also a little quiet but

we hear you so um what were

the what were the themes that that uh

connected the

presentations that you were in or

otherwise one thing that um struck me

can you hear me all right now is this y

excellent I mean there are lots of lots

of little uh golden nuggets um

throughout all the presentations um but

one um one crosscutting theme was this

notion of community that emerged in

various guises such as the notion of

Commons that was the subject of uh a lot

of the exchanges in the first um in the

first

um uh

presentation um and then the notion of

community that was a Dumont of the the

workshop um and then the not of
coherence and synchronization and
narrative social um science accounts of
that and how that might be formalized in
terms of um joint free energy
minimization ensembles of Agents you
know finding a coherent communal um and
empowered free energy Minima so you know
I thought I I thought that that was a
common theme sort of
focusing much less upon um
individual agents but what happens when
you have a an ecosystem of Agents um and
what
principles um can be brought to the
table to understand interactions or
ensembles of Agents so I I have a number
of you know lots of things came to mind
I was trying as a physicist to sort of
um rewrite all of these wonderful
Concepts um in a mathematical way or or
a physics way um so for examp examp I
was think about the importance of
Commons and it struck me that one way
that we could motivate the choice of
capital c for prior preferences uh would
be that these are the common goals and
the and the C characteristics States um
that are shared between um multiple
agents uh and of course um the last
presentation took us through some of the
mechanisms and the capital c circular
capital c
causality um that leads to this Capital
 C coherence so you know which originally
was meant to be constraints in the sense
of a constraint Maxim entry principle um
or cost in the you in a reinforcement

learning uh reinforcement learning
setting but coming back of course all
this started when I was hearing about
Commons and how it underwrote
communities it is very interesting
how development and the presentations
have
proceeded a lot more about the
communication and somewhat of the
multi-agent coordination as opposed to a
longer memory train for individuals or
deeper prospection for one individual
um and in a way the
renormalization method brought a a
Clarity or a rigor to what had already
been qualitatively
explored from 2013 and Life as We Know
It And all these other works that we're
talking about multiscale systems and
about blankets of blankets and within
this year to make it timely we start to
see some of the
ways in which that compositionality can
actually be brought into the generative
models that we've seen I mean it looks
like the textbook figure 4.3 and the
nested models and now that's
actually the
motif that gives a way of looking at a
multiplicity at the more inactive level
as having a Unity at a higher level
Alex in
your designing or work how how does the
single agent or
multi-agent design come into
play
now yeah yeah can hear okay good yeah I
was having mic troubles earlier um well

so it's kind of interesting I think I've spent most of my career at the agent level in the sense that I want to build I don't think we have right the single agent and I and I think it's kind of important to sort of scope that out from an architectural learning perspective I'm very much a fan of like Carl's arguments and you know I Echo them myself a lot which is the idea of the inference learning and structural time scales uh I still think we have a lot to do but if I were to say how in my research the multi-agent approach really comes into play well I do think a lot in terms of assemblies or ensembles right putting together multiple circuits I'm a I'm a huge proponent of another type of this comes from cognitive science called the common model of cognition uh I'm not sure how many in the audience are familiar with it but it's sort of like a generic reprint that has emerged over the last several decades of cognitive science to say that we can think about and take our current Knowledge from neurop physiology and cognitive studies that brain has these rough levels of organization different types of memory we have the motor cortex we have phas of gangway we have all these very important structures and we can start appending to them some functionality and so in that sense that's where I think of multiple agents right because you can start breaking apart using again Carl and I've have argued this the mean field approximation

uh there's natural mean field approximations that emerge through the result of union where you have these little substructures that communicate with each other um obviously the sparse connectivity that Carl mentioned and his talk this morning is uh kind of indicative of how these little substructures emerge and then communicate and perform sort of like what uh Marvin Minsky would have said you know sort of a society of mind right uh so it's kind of a call back to that and I think in that regard I often uh one branch of my work often tries to highlight and emphasize bring forth that notion of cognitive Blueprinting or cognitive structure um because then we can start filling in the pieces with what we know about neurobiology like for example I use predictive coding quite a bit uh you know to fill in those gaps and so I think in that sense I think a lot of interacting subsystems you could think of those as agents and then my final comment uh without letting myself blab too much uh I have worked in neuroevolution so there is another scale Above This which is you know you do need to account for how multiple mortal computers or multiple agents interact with each other um and so I think in that regard I have done some work with very simplified agents where you can kind of abstract away the neuroscience and just say I want to look at these little tiny systems that interact and

then we can you know modify them through genetic algorithms and neuroevolution so there is a little bit of that like higher level uh societal kind of learning I have looked at that and I think a big challenge just to bring it to the audience for the panel is uh merging those time scales in a in a coherent and efficient way I still think well there has been good work I know Daniel you work on and Colony optimization as well and other types of kind of aggregate algorithms uh I think bringing those time scales together and really interacting with what we currently know about single agents even like the common model of cognition and going back to the highest level which is this multi-agent evolutionary time scale is a Grand Challenge and how do we of course effectively simulate those in a way that anyone can play with this I think is a big Challenge and yeah and I think I'll kind of stop there before I go down more radicals neuroevolution and neural Darwinism something Carl and and others have also explored it's like they're all and then the multiple nested time scales multiple nested spatial scales it's like there's this attractor to want to come back and find the unifying patterns and and it's and how those can be applied is kind of an exciting question I'll read one question in the chat so feel free to give a thought or take it any direction David Highland

wrote do you view mortal and IM do you
view mortal and Immortal computation as
binary categories or are they more like
two points along a
spectrum actually want Carl to answer
that I mean I could give something but I
really want to hear your thoughts on
that it's your favorite subject though
design I know but I'm actually really
curious if you think of it as a spect
because remember the conversation we've
had with Mike Levan and Chris fields and
we've been sort of debating you know how
important it is to lean into mortality
versus immortality do you think that
there could possibly be anything in
between because I honestly am you know I
think of it kind of more rigidly I'll
just give my answers that and I firmly
think we need to like focus on Mortal
computation but do you think that
they're can be a hybrid in between so
sorry to turn it on you but I'm very
curious to know your thoughts um my my
instinctual answer is no
um having said that I am sure that you
can present some um description of
mortal computation in terms of immortal
algorithms I mean I'm just thinking back
to sort of um um the trilogy due to
David Mah uh but does that mean that
these kinds of descriptions are apt um
for simulating reproducing
understanding um or describing the
actual process I I I suspect not they
may at one level of description they may
be but when it comes to
um applying the principles usually in

computer simulations I suspect that that
just would not work I
mean because Daniel and I were wrestling
with zoom and other um um communication
things I I I may we may have missed
whether you've taken people through the
the the key distinction between Mortal
and Immortal uh computation so AI do you
just want to briefly rehearse the sort
of the two Cardinal views of the of the
distinction um between Mortal and
Immortal uh computation I did I just
wasn't sure if anyone heard me so I
wasn't sure what was going on
Zoom yeah I mean certainly from the
point of view the free energy principle
the free energy principle is a
description of a random dynamical system
um
and as such um it is
um a system that constitutes the world
the you know the lived World um perhaps
not the lived world but certainly a
mortal world um that has um
characteristic time scales and therefore
um the free energy principle and
everything that ensues from that
including active inference um rests upon
that primary assumption that you are
describing a process a mortal process a
a a process that could have um a
physics um although
um the actual free energy principle
itself is almost pref physics so you
have to derive basic mechanics and
classical mechanics and quantum
mechanics from um the existence the your
the the the mortality of this process

cast in terms of your time involving
systems so in that sense um you know
there is no such thing as Immortal
computation
but I also I don't know if you said it
out loud but I also liked your um more
playful take on Mortal versus Immortal
that if so if software was
Immortal uh and it had some autopoietic
aspect to it then it wouldn't care about
dying so there would be no wait there'
be no way there' be no incentive to have
characteristic States coming back to
another big sea um uh so you know in
that sense they cannot be immortal
computation simply because that is
removing the definitional uh stipulative
definitional aspect of the
characteristic states that we are trying
to describe through the Fens principle
and active inference and specific um VAR
variance of that such as predictive
coding but I just pick up on this
crosscutting CC theme uh you also um um
celebrated that with common cognitive
models so again with we've got this this
s notion commonality and um and you
asked the question you how How would how
would one um apply these Notions and if
you like sort of naturalize the common
cognitive model under um a less
anthropomorphic or functionalist active
inference account and you gave the
answer implicit in your question which
how would you simulate these things um I
think Daniel asked the same the same
question indeed the the whole point of
this Symposium is about applications so

I I just wanted to put a question to the two of you I was um impressed with the very um clear distinction between agent-based modeling and the common cognitive equivalent the where the agents the members of The Ensemble were actually for example active inference agents that seems to me a very big move and a very difficult thing to actually do um most of the literature I know in this area um has at most taken baby steps into having ensembles of two agents with authentic agency in the sense that they can plan so I'm now limiting um the notion of an agent Beyond a thermostat or a virus to those kinds of systems that um have in mind or part of their Gena model the consequences of their Lear actions so they can be future pointing and they can plan and select from alternative policies uh and of course the numerics and the you know the the Practical difficulties of actually simulating doing agent-based modeling for example um with authentic agents in the mix in the ecosystem um I think is you know um is a really important Challenge and a really important distinction between classical agent-based models where you put lots of ther stats or rents tractors together and see what happens and of course they have a synchronization manifold and you can certainly wax lyrical about um generalized synchrony and and and Joint free energy minimization of a generic sort but I think that misses the point I

think Ma was making um
um that you know part of the the
multiple L Loops of circular causality
in sort of your know cultural Niche
construction for example um you know
really does rest upon um the capacity of
any individual within an ensemble um to
um act given they have selected a
particular course of action particular
path into the future which they have
beliefs about the consequences of and
you know to me that that that that is an
authentic kind of agent so you know mean
perhaps well both of you d mean your
your your ethology background um are you
aware of anybody's actually sort of
Bitten the bullet that was introduced to
US during the workshop and actually
started creating active inference agents
that actually plan and learn from each
other and carve out their own Wallington
Landscapes you know finding those sort
of joint joint

Minima I don't know specific uh citation
but certainly you've given a research
Direction and it reminded me of in the
shared protentions work where thinking
about what what does a shared Ensemble
what would it be for a planning Ensemble
or for a collective it could be a shared
protention as in there's the same plan
disseminated across the ant nestmates it
could be that they have different maybe
nonoverlapping but complimentary parts
that would be shared it could be having
a shared reference but they could have
different plans or it could even be a
shared consequence without even a shared

reference so there's so many ways in which especially when we consider ensembles that that are heterogenous and they have different kinds of umelt different kinds of ways of engaging with the niche and and possibly some comment niche as well as some uniquely accessible components there's so many flavors of

Ensemble that I I can can only hopelessly or helpfully try to think about what tools and infrastructure and coordination would even give us a grasp to be able to sweep or design or visualize across that otherwise we might be locked in

way too early to what it means for The Ensemble to have a collective protention and then that's not going to apply to every ant species and if it can't even apply to all the ants out there then it's not going to work for

aliens Alex can I can I just respond to that and then ask ask for your your view

on that one so I think it's a really important point and um one of the big takeaways for me on our paper on the variational synthesis you you're applying the renormalization group to the the coupling of evolutionary to um sort of phenotypic uh time scales was that it was not uh feasible or

appropriate to talk about consp

specifics in isolation that it was you

had to have the all the varieties of

predators and praise and things that we

um that not even animate not even agents

in the mix to actually
um derive the um your the
non-equilibrium steady States at
particular at particular scales which
you know speaks directly it is not um it
is not uh useful to um you know if you
like take the idealized gas mentality of
20th century physics and now apply it to
agent-based modeling to assume that all
the agents are exchangeable and are
identical in some s sense and then are
converging on some common narrative um
or some common um uh is not the right
way to you know to simulate these things
you can't do Predator prey relationships
you know for example you can't have um
even simple economic games uh but
furthermore you you you when people talk
about depletion of resources you
actually have to model the resource you
know whether it's oxygen food or water
as things in in that ecosystem so I
think that's a really important
observation which just speaks to the
challenge of using numerical studies to
test out some of the hypotheses that uh
that we' you know being being exposed to
um you
know you know if one just pursues
agent-based modeling without introducing
that heterogeneity I think one is making
exactly the same mistake um that people
make when they apply uh equilibrium
physics to non-equilibrium systems um
you know you know
a true gas is not an ideal gas and it
only has the kinds of properties and
breaks detailed balance and expresses

non-ra in virtue of the fact that there is no exchangeability you can't assume that every atom or every agent in our context um is exchangeable with with every other agent and to connect again to Commons and and that model of well within the firm we assume Co cooperativity and hierarchy in the market we have a flatness and a competitiveness and then there's this Commons that has sort of elements of each so then I think about doing ecosystem modeling it'd be like well we have within the ant colony we assume cooperativity possibly hierarchy among ant colonies you have red and tooth and Claw and competition and then it's like but can there be a unifying framework that would have the two genotypes and two phenotypes and two species and two adjacent niches and keep on adding in these levels of Distinction and seeing all those different variables in their joint Evolution across multiple time scales rather than the concatenation of a Cooperative hierarchical game NE Within in a lateral competitive game it's like capturing what those models get the the width and the height of but bringing it into a framework where it's being seen as just one joint evolving process so C is the letter of the day uh Alex can you speak to CES of cells what what that kind of heterogeneity and structured diversity would look like

from a point of view of tissue for
example or an organ okay wait what was
the c word you said I just didn't hear
it it's any any word starting with a
letter C you can use but you have to oh
I have to use okay this is the
game um Community I'm gonna pick that
one just because it starts with c um so
okay Carl threw in organs and organel
but I'm going to push that aside for
just a minute to address I think one of
the questions I heard emerge from yours
and Carl's discussion which was uh is
there anything like what you are arguing
for which is avoiding the mistakes and
by the way Predator prey model came
immediately to your speaking Carl um
about these classical models that
assumed interchangeable agents and is
there anything that looks at these
heterogenous systems or heterogenous
agent interactions and a couple of
thoughts I don't think there's anything
like directly what we would want and I
think that ties or centers around our
even the arguments you and I and many
others in the community are making about
uh why it's difficult to show case why
active inference is so important even
for places like machine intelligence um
but I did write some thoughts quickly uh
so I do think uh the first thing that
came to mind weirdly enough was an
application in the world of machine
intelligence where they had some
environment where they had a community
of even though I say it with slight
distaste large language models or Chad

gpts or gpts that had some extra modality equipped to them and they had this like little Community now it was a virtual very simplified Niche if you will um but the idea is that they could observe this community um and so these different gpts would and I can try to dig up the paper and send it to Daniel later it's one of my students that actually pointed it out to me they would different roles they had different personalities if you will um and I know that they investigated that to some degree now of course that doesn't have any active inference in there and I'm not sure what degree of maybe perhaps light reinforcement learning being introduced but there is something like that and there is slight baby steps if you will uh in investigating some notion of embodied multi-mod and agent systems I think the AI Community is still sort of barely starting to embrace embodiment let alone enactivism and let alone embeddedness which is the idea of a community to go back to the sword of multiple agents interacting with each other um and investigating how do social norms and merge or cultural norms and cultural constraints and relationships at this like more Global level so I think there is a little bit of those Sparks of that thought and obviously I might not be aware of further follow-ups another place that uh I at least see the potential of no longer having these interchangeable particle like agents is in multi-agent robotics um I know that

they have done things like soccer competitions and they have these little and again each of those robots they are the embodiment of what you and I want which is a version of a mortal computer so it's a little like assembly of mortal computers and there's a Cooperative nature so and then a competitive nature so you get a little bit of a very constrained real physical real world Niche um obviously it's in the constraints of it's a soccer game or some very simple task so I know in multi-agent robotics there is some development along this path and it'd be wonderful to have like a a a hardcore because I'm more of a simulation roboticist a hardcore roboticist to talk to us about the state of multi-agent Robotics I think that's the place where you could begin to test uh hypotheses and kind of or theories that are constructed even from active inference point of view uh into these real world robotic multi- assemblies you know they have like we could analyze the competition Cooperative Dynamics I think that's our best bet in terms of what maybe exists at the closest to literal agents that aren't just very simplified even though they make simplifications in robotics too uh in order to have a little brain in there that you know does something meaningful usually learning is turned off which is my own criticism of Robotics um and I think the last comment I wanted to make and again getting back to the Notions of what we need so again

you and I have a fondness I think for some of cybernetics and cybernetics has this notion of complex emergent systems and we you know you have these little uh building blocks at one level that follow their own physics or their own physical laws and then they maybe assemble or emerge into these higher level building blocks that are subject to their own laws then you have bottom up in top down causation which is a notion that you know comes from cybernetics and we always talk about that but then I think about like Santa you you might be familiar with the Santa Fe Institute and all their wonderful work like uh what was it not just Herbert Simon but uh uh John Holland and all the wonderful work on complex adaptive systems but they always did at the end of the day when they simulated it they did the particle thing this interchangeable particle simulation and then tried to analyze like those convergence properties or those emergent rhythms or patterns that kind of emerge in those systems with no to my knowledge and you know and I did study complex adaptive systems a while ago uh not with that heterogeneity and let alone any of the complex agents that we could begin to design today even with deep learning regardless of its limitations and so I think that that maybe even turns into an answer to your question highlights an open opportunity Maybe revisiting and recasting Notions of cybernetics and complex adaptive systems in this idea of well can we

actually build more complex agents uh
for example that actually have neural
substrate structures or something that
we can simulate and then learn to
acquire their own characteristics in
their Niche I think the reason that
maybe a lot of people avoid it or maybe
why I don't perceive uh development in
that direction is the problem also of
Niche and you and I also had an and a
not to try to highlight this paper but
in the appendix we talked about the body
Niche problem uh and that is this issue
that uh you need a niche if you don't
have a niche and you're designing these
agents independently like deep learning
typically does to apply to any Niche uh
you're missing really half the problem
which is understanding the physical laws
and how do we simulate them such that
people can then play with those models
maybe you don't have access to a xenobot
like Michael Levin does or I don't have
access to organoids though I wish I did
um but you know maybe we can simulate
these and so I think that we need the
community may this is a call to the
community to develop uh viable niches
virtual morphologies virtual uh areas
that we can then connect these agents
and then simulate many of them and try
to analyze and understand do the things
we predict even with these
interchangeable particle models apply to
these heterogeneous communities that
might emerge and again I guess that just
to wrap up and I promise I'm done you
brought up organelles and organs and I

think that highlights something that I wish I had more easy access to rather than uh being put into this position where I well may I need to build it which is you know these virtual organ systems or virtual organ organel and trying to understand how these dynamical systems interact and then I can kind of take that as that's my body or that's where my Mark and I can kind of start to put some instantiation of blankets and then say Okay I want to instill maybe a neural circuit maybe I put in a predictive coding model into this kind of virtual morphology so I can analyze and understand does do these interactions of the time skills actually come through when we actually build you know physical instantiations of these models and so I think that's going to be one of the biggest roadblocks to even my dreams of mortal computation which is this notion of myology something that we can actually work with that it gets as close as we can to reality without reality and in Niche um because once you have those problems solved then you can start building multi and if you have efficient way to simulate them and I'm sure lots of people that are watching might be worried uh because it's very hard to simulate you know real physical systems let alone allow machine intelligence researchers to play with them so I blabbed a little bit there but does that sort of give you some something to bite on based on your questions something to chew on something

to consider um in the last little bit
I'll I'll ask some questions so I'm
going to read one from the live chat and
then anyone else we'll just do some
quick fun random questions sort of like
sampling from the uncertainties
expressed uncertainties of our live chat
colleagues okay Pablo FM wrote is
playing a simulation acquiring knowledge
SL experiences through simulation of
scenarios
and how can we use active inference to
do
games how do you fellows make your
active inference work playful and fun if
or
when let you go Carl you go I'm not sure
if there are two there there are two
questions there
um a practical answer to um why you
might want to
instantiate um a simulation environment
with active inference agents or indeed
just a single agent in the in the spirit
of computational phenotyping but I think
probably more interesting in the um in
the context of um communities of Agents
um is in the spirit of this sort of
cognitive um agent based modeling um is
the ability to um parameterize the
priors of the agents to best explain or
replicate Legacy data from a particular
ecosystem be it Financial
epidemiological
meteorological
um purely ethological
um and having optimized the parameters
you then roll out into the future

and in rolling out into the future you now have the opportunity to have a grounded in principle base optimal plus inactiv ISM and embedded um um estimate of what is likely to happen in the future again in the spirit of weather forecasting but of course this becomes much more um if like useful um when you have control of the system so Financial Market forecasting or epidemiological forecasting or uh climate change forecasting you assuming that there's a substantive um an um anthropomorphic um um anthropological um contribution to um climate and you all the other factors that uh you know um pertain to um resilience and uh resistance to things he you know health and the like um as soon as you can intervene then once you've got a digital twin of this ecosystem that has been optimized in relation to um the past data in the sense that this now has the highest model evidence or marginal likelihood in relation to that Legacy data you've then got an opportunity to do scenario modeling with interventions so you can now ask questions what would happen if we um cut this Supply CH train chain what would happen if we impose this lockdown in this viral pandemic what would happen and so on and so forth furthermore because you've got um um uncertainty in the mix because this is effectively going to be um you're using variation procedures that we use in active inference to actually simulate um

the behavior that best matches the um
the Legacy data you've also got for free
uncertainty quantifications and not only
can you say if I interv on this system
and did this um then we would uh I can
predict that in six months time or in
six years time that and I can give you
Bas in credibil intervals o over that
moreover you can now apply the
principles of active inference to roll
out under different interventions which
now become policies and evaluate the
expected free energy of each policy and
select the one under the constraints the
Comm the characteristic outcomes U that
you um commit to at this point in time
and you can now use this scheme to make
recommendations on the basis of the um
the relative probability of this
sequence of interventions versus that
sequence of interventions so I think
there are great practical importances of
being able to put these principles in
silico and basically do Evolution very
very quickly so that tomorrow you can
read the you can read the consequences
probabilistically uh of evolution in
this setting and that setting where the
setting is actually something that we
can intervene on or even if we can't
just to know the consequences of our
Collective Behavior Uh today so I think
that would be um that's why you might
want to make you know to invest in in
these applications um with the fun and
uh having uh and play I think that's a
different issue I think that's basically
um um

one way of looking at the um the way
that authentic agents behave under under
the free energy principle which is
basically to explore and gather the
right kind of evidence that uh provides
literally evidence for their generative
models of of their world um which
basically entails novelty seeking but
now complement this with this um basing
reduction we've been talking about also
trying to find the minimal explanation
for the data and the evidence that has
been secured by exploratory playful
Behavior arriving at particular insights
which uh those are har moments that say
oh I see it's just one of those or this
one thing explains all of this playfully
AC acquired data and perhaps that kind
of mechanism you know may be a really
important aspect of the communal um um
you cognitive agent based or yeah
cognitive agent based I can't quite
remember the acronym it was very nicely
laid out in one of the I have to go back
and look at the live
stream a follow-up question from the
live chat Arno wrote I agree with the
approach to calibrate to pass data and
forecast but it's already quite
challenging for current models that only
have a few parameters to be calibrated
how then does one calibrate prior
beliefs of millions of
Agents I'm talking about epidemiology
and economic models sorry I'm going to
give a technical answer before Alex
gives a gives aedy answer uh you do it
it's only difficult because 99% of

agent-based modeling uh uses the sample distribution um you do not do that you use variational you actually um simulate and roll out the um the um probability distributions themselves using variational calculus and that means that what was yesterday um really difficult agent-based modeling and certainly epidemiological modeling now becomes really trivial uh orders of magnitude more uh more efficient what you have to take care of is exactly the thing that Daniel was highlighting before which is a heterogeneity you need to put that into the mix and this manifests in a number of ways um so when you apply um variational procedures to effectively um roll out population density uh probability density distributions over various states of being am I infected am I staying at home um have I died um you have to make sure that the heterogeneity is part of the generative model so this is sometimes called stratification so for example in epidemiological models you'd have eight bands for different age groups and if you notice the way that the epidemiological data was generated was usually age segregated so that people could stratify their models um another um way in which this heterogeneity becomes absolutely crucial um when you come to um agent-based modeling in the context of epidemiological or infectious disease modeling is called over dispersion now over dispersion is the fact that uh you get heavy tailed distributions that

inherit from the heterogeneity so you get this for free once you put heterogeneity into the model but it does mean um for every if you like subgroup of um the population you do induce more parameters but remember the number of parameters it doesn't really matter it's the amount of time that you take to effectively invert your model and if you try and do that using sampling based approaches which is currently the the norm you could take literally weeks to uh do something um whereas if you use variational procedures then you could do that within within minutes so it's a really important question a very practical

one yeah to restate direct variational inference on beliefs about ensembles themselves being variational belief inference on agent level semantic beliefs rather than sample from the sample what the agents are sampling yeah

you're you're you're invoking the The Twist yes how on Earth can you model a variational density over things that have variational densities you can do it under the f p room because the only observable things about an agent are on its Markoff blanket and of course they are not belief structures so there is a lawful relationship between you can um if you like simulate cognitive agents using traditional agent-based uh approaches but what you have to do is to amortize or to or to um um basically work out the um the mapping between the

history of what something sees um and
what it actually does but in principle
you can do it by marginalizing out the
um the beliefs of the agent in principle
I'm not and that's that's why I asked
that's why I'm so intrigued by this
distinction because that will be a
challenge Alex and then Carl just in our
last minutes here what would you say was
a favorite memory or moment in theory
and practice for active inference this
year and then what are you excited about
for next
year um I'll go I'll go first so a
favorite memory of active inference let
me think quickly well I I mean I'm going
to be a little by
just to give a highlight and a shout out
and everyone will get to share in that
memory if you stick around after the
round table is actually some of the
results from one of my PhD students um
via and he's going to give a talk soon
and uh the idea is that uh one of the
things I was proud of is he's been
working in the area of Robotics which I
think is a really good playground uh is
difficult and hard is it to democratize
for everyone to work with robotics I
think it really shows you how far we
need to go and I think that's a good
place to make strong advances an active
inference um and one of my memories is
that I remember him kind of showing me
uh because he's working with partially
observable Markoff decision processes
and how difficult and
unaddressed it is in the field of

working with raw sensory data to try to construct these continuous models and I remember him showing me some of his models rolled out trajectories in the future and I I'm always kind of in a fondest in my heart for generative models and I remember him playing out sort of these robotic memories of this robot trying to you know fantasize where it would want to go to kind of like move this block or to knock a ball towards a go so that was one of those moments where we were really proud because the idea is we actually we actually made a breakthrough uh in actually showing that we can construct a very robust framework that applied across all these robotic control problem and so I was immensely proud of that and so I think my students drive a lot of some of my fond memories um and uh and then we got uh Chris Buckley shout out to him uh to you know uh kind of give us his insights into our work too and it was a really really nice work and so and I think the best way to say it is stick around if you want to know how that all works I was really proud of that so one of my favorite memories of the Year awesome Carl with the last words of a fun memory and uh a direction or an excitement for next year right I'm I'm not going to tell you about my most fun moments that's that's very personal but my my my my excitement for next year is to see whether any of your hairs go gray after you do the master ceremony of these of not only

these uh Symposium all the meetings
because for those people who don't know
Daniel has to run this almost
singlehandedly nonstop for a marathon
sometimes 10 hours at a time I don't
know how he does it at Le unless you're
using hair dye you're you're you're
standing up physically remarkably well
sir so I just see how long that
lasts thank you Carl I look forward to
the marathon
changing continuing and flourishing in
the coming
year we appreciate you Daniel thank you
guys it's been
awesome okay thanks a lot see you soon
nice talking to you guys bye
everyone
e e